

Balancing the Dose: RL Agents for Sepsis Treatment in the ICU

Reinforcement Learning Project

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1. Introduction

Every year, sepsis claims millions of lives in intensive care units across the world. What makes it particularly difficult to treat is not a lack of interventions, but the challenge of knowing when and how aggressively to apply them. Two decisions appear throughout every patient's stay: vasopressor dosage to maintain blood pressure, and intravenous fluid volume to sustain circulation. Too little administration causes deterioration; administer too much and the treatment itself is the harm. The optimal balance shifts continuously as the patient's condition evolves.

Traditionally, clinicians navigate this complexity through protocol and experience, but the decision space is vast (hundreds of distinct patient states, dozens of possible treatment combinations, delayed and sparse outcome signals, and high individual variability). These are precisely the conditions where Reinforcement Learning excels: an agent that interacts repeatedly with an environment, receives delayed feedback, and gradually refines its policy without requiring labelled demonstrations of correct behaviour. This project explores that possibility using ICU-Sepsis-v2, a benchmark environment, in which each episode replays a real patient trajectory.

The problem is approached in two stages of increasing complexity. In Configuration A, the full MDP transition model is available and patient states are discretised into 716 clusters.

This controlled setting allows a direct comparison between two distinct algorithms: Policy Iteration, which exploits the known dynamics through exact Bellman evaluation; and Q-Learning, which must discover the same optimal policy through interaction alone, while isolating the cost of model access. In Configuration B, a 47-dimensional vector of raw physiological measurements makes tabular methods unfeasible. Three wrappers further stress-test the agents under realistic deployment conditions: sensor noise, missing laboratory values, and sudden irreversible acute events. Here we compare a Deep Q-network (DQN) with Double DQN updates against Dueling DQN, whose separated value and advantage streams are hypothesised to be better suited to states where the choice of treatment is less important than patient severity.

Beyond survival rates, we argue that a clinically trustworthy agent must also demonstrate safe treatment behaviour. A creative extension therefore analyses the treatment strategies each algorithm develops across the vasopressor-fluid combinations grid, connecting learned action preferences back to the foundational clinical principle of *primum non nocere* - first, do no harm.

2. Methodology

2.1. Problem Formulation

The sepsis treatment task is formalised as a Markov Decision Process (S, A, P, R, γ) , where S is the patient state space, A the set of 25 discrete treatment actions (5 vasopressor x 5 IV fluid levels), P the transition dynamics, and R the reward function. The reward is sparse and terminal: +1 upon survival, 0 upon death, with a per-step intensity penalty $\lambda = 0.02$ discouraging unnecessary intervention. The discount factor $\gamma = 1.0$ is fixed by the environment, treating all timesteps as equally important.

The two configurations differ in how S is represented and whether P is accessible: Config A exposes a discrete state index (0-715) alongside the full model; Config B replaces this with a 47-dimensional continuous observation vector, making P unknown and tabular methods unfeasible. Config B additionally applies three wrappers that inject real ICU deployment failures into each episode: *EpisodicNoisyObsEnv*, gaussian sensor noise corrupting the full observation; *EpisodicMissingObsEnv*, selected features zeroed out, simulating unavailable laboratory values; *AcuteEventEnv*, irreversible patient deterioration independent of treatment, simulating cardiac arrest or acute organ failure.

Figure 1 illustrates the reward sparsity of the environment: for the vast majority of states, the maximum achievable reward across all actions approaches zero, confirming that meaningful feedback is confined to terminal transitions and making credit assignment particularly challenging for model-free methods.

2.2. Config A — Tabular Methods

Both Config A algorithms target the Bellman optimality objective but differ in whether they exploit the available model. This was deliberate, to allow a controlled measurement of what model access is worth in terms of final policy quality and convergence speed.

2.2.1. Policy Iteration

Policy Iteration alternates between two steps: Policy evaluation computes the exact state-value function for the current policy by sweeping through all states until convergence, and Policy improvement updates it greedily with respect to those values.

It converges to the exact optimal policy in a finite number of sweeps, without environment interaction or exploration, using a convergence threshold of $1e^{-6}$, a value small enough to ensure precision without unnecessary computation. Its solution serves as a reference against which the model-free agent is measured.

2.2.2. Q-Learning

Q-Learning estimates the optimal Q^* from sampled transitions, bootstrapping on the maximum estimated future value:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

independently of the exploratory behaviour policy, using an ϵ -greedy schedule from 1.0 to 0.01 over 30,000 episodes via a decay rate of 0.9997. Additionally, a learning rate of 0.1 balances convergence speed and stability on stochastic transitions. Under GLIE conditions, convergence is guaranteed in the tabular setting, making comparison with Policy Iteration theoretically sound.

2.3. Config B — Deep RL

With a 47-dimensional continuous observation, a Q-table is no longer feasible, being replaced with a neural network trained via stochastic gradient descent to minimise the mean-squared Bellman error. Two stabilisation mechanisms are shared across both architectures: Experience Replay, storing transitions in a buffer of capacity 50,000 and randomly sampling in batches of 64, breaking temporal correlations that would otherwise destabilise gradient updates; Fixed Target Network, a frozen copy that computes TD targets, synchronized with the online network every 100 steps, preventing the moving-target problem.

2.3.1. DQN with Double DQN updates

DQN extends the standard Q-network with a single modification to the update target: we replaced the max target of standard DQN loss with Double DQN updates, decoupling action selection (online network) from action evaluation (target network):

$$y = r + \gamma Q(s', \arg_a \max Q(s', a'; w); w^-)$$

This decision was made with the intent of reducing the maximization bias, which is amplified in this configuration by observation noise from the clinical wrappers (noisy inputs produce noisy Q-estimates, and the standard max operator systematically selects the most overestimated action).

2.3.2. Dueling DQN

The final chosen algorithm was the Dueling DQN, which modifies the network architecture to decompose Q into a state-value stream, and an action-advantage stream:

$$Q(s, a) = V(s) + (A(s, a) - \frac{1}{|A|} \sum_{a'} A(s, a'))$$

This combination ensures identifiability, preventing V and A from shifting by an arbitrary constant. The clinical motivation is direct: some sepsis states are dominated by severity rather than treatment choice - a patient in irreversible multi-organ failure will likely die regardless of the next IV fluid dose. In such cases, the value stream learns efficiently without waiting to visit every action, accelerating learning across the state space. The advantage stream then captures the marginal benefit of specific interventions in states where treatment does matter.

2.4. Hyperparameter Summary

Hyperparameter	Policy Iteration	Q-Learning	DQN	Dueling DQN
γ (discount)	1.0			
$\frac{\theta}{\alpha}$ (convergence/lr)	$1e^{-6}$	0.1	$5e^{-4}$	
ϵ start $\rightarrow$ end	-	$1.0 \rightarrow 0.01$	$1.0 \rightarrow 0.05$	
ϵ decay	-	$\frac{0.9997}{episode}$	exponential	
Training Episodes	4 iterations	30,000	10,000	
Replay Buffer	-		50,000	
Batch Size	-		64	
Target Update (steps)	-		100	
Hidden Units	-		256	
Network Depth	-		2 layers	2 + 2 streams
Optimiser	-		Adam	
Grad Clip	-		10.0	

Table 1 - Summary of hyperparameters

3. Evaluation

3.1. Config A

All results are evaluated over 1,000 episodes using a fixed greedy policy ($\arg\max Q(s)$ with no exploration) under a fixed seed for reproducibility. Survival rate is defined as the percentage of episodes in which the agent receives a positive return.

3.1.1. Policy Iteration Convergence

Policy Iteration converged in 4 iterations, consistent with the theoretical expectation that dynamic programming methods require only a small number of evaluation-improvement cycles on well-conditioned tabular MDPs. Figure 2 shows the maximum value change per iteration on a logarithmic scale, decreasing from 1.0 at the start to below 10^{-3} by the final iteration - an exponential decay reflecting the contractive nature of the Bellman operator. The greedy policy stabilised after the first improvement step and did not change in subsequent iterations, confirming that the optimal policy was identified by the algorithm well before the value function fully converged.

The final greedy policy achieves a survival rate of 78.8% and a mean return of 0.7507, representing an improvement of +8.5 percentage points over the random baseline (70.3%). The mean episode length decreases slightly from 10.4 (random) to 9.8 (PI), suggesting the optimal policy reaches terminal states more directly, without requiring exploratory detours.

3.1.2. Q-Learning Training & Convergence

Q-Learning achieved near-greedy behaviour around episode 19,000. Figure 2 shows the smoothed training return alongside the raw per-episode signal. The high variance in early training is expected: with ϵ near 1.0, most actions are random and the sparse reward provides infrequent learning signal. The smoothed return stabilises around episode 15,000, after which further training produces marginal improvement, suggesting the agent has effectively converged within the allocated episode budget.

The final greedy policy achieves a survival rate of 68.5% and a mean return of 0.6029 - marginally above the random baseline in return terms but 1.8 percentage points below it in survival rate (70.3%).

This result warrants careful interpretation. The survival gap does not indicate that Q-Learning is worse than random policy execution, since the random policy is evaluated without the intensity penalty, while Q-Learning's return reflects a policy that has partially learned to avoid unnecessary interventions. More importantly, 30,000 episodes with a sparse reward distributed across 17,900 (716×25) state-action pairs leaves many entries undervisited. Figure 5 shows the Q-table heatmap: while Policy Iteration produces dense, high-value entries across all state-action pairs, Q-Learning's table contains large regions of unvisited, near-zero values, directly evidencing incomplete coverage.

3.1.3. Config A Comparative Analysis

Table 2 summarises the final performance of both algorithms against the random baseline.

Algorithm	Mean Return	Survival Rate (%)	Avg Episode Length
Random baseline	0.5937	69.7	10.35
Policy Iteration	0.7507	78.8	9.81
Q-Learning	0.6029	68.5	9.9

Table 2 - Config A performance metrics

The performance gap between the two applied algorithms - 10.3 percentage points in survival rate - is attributable entirely to model access. Both target the same Bellman optimality objective; Policy Iteration solves it exactly using the known transition dynamics, while Q-Learning approximates it from sampled experience. Figure 7 illustrates this directly: the state value function computed by PI is consistently higher and smoother across the 714 non-terminal states, while Q-Learning's value estimates show high variance and frequent drops to zero in states that were rarely visited during training.

The policy agreement analysis (Figure 6) quantifies this divergence: the greedy policies of the two agents select the same action in only 8.3% of states. This low agreement is clinically significant - it means that a deployed Q-Learning agent would recommend substantially different treatments from the optimal PI policy in over 90% of patient states. In a clinical setting, this degree of policy divergence would represent an unacceptable deployment risk, reinforcing the importance of either sufficient training data or model access when the action space has direct patient safety implications.

The convergence comparison (Figure 2) further highlights the sample efficiency difference: Policy Iteration reached its optimal policy in 4 iterations, almost instantaneously, whereas Q-Learning required 30,000 episodes and still did not fully recover the optimal policy. This result is consistent with the DP limitations discussed in Section 2: without the model, every state-action pair must be visited and updated individually, and sparse reward slows this process considerably.

3.2. Config B

All results are evaluated over 500 episodes - compared to 1,000 in Config A - using a greedy policy derived from the trained neural network, under a fixed seed. The reduced episode count reflects the higher computational cost of the clinical environment wrappers. Survival rate is defined identically to Config A. Both agents were trained for 10,000 episodes in the clinical environment with all three wrappers active at the same time.

3.2.1. DQN Training & Convergence

Figure 8 shows the smoothed training return for DQN over 10,000 episodes. Unlike Config A, where Q-Learning exhibited a clear convergence trend, DQN's training return oscillates throughout without fully stabilizing, a pattern characteristic of deep RL on noisy, partially observable environments. The high variance is attributable to the clinical wrappers: episodic sensor noise and missing features mean the agent may observe considerably different representations of the same underlying patient state across episodes, destabilizing gradient updates despite the fixed target network and experience replay.

Survival rate evolution (Figure 9) follows a similar pattern, oscillating above and below the random baseline without settling into a stable plateau. The TD loss curve (Figure 10) shows an initial spike followed by a gradual decrease, confirming the network is learning.

However, the persistent oscillation in survival rate suggests the policy itself still remains unstable throughout training. The final greedy policy achieves a survival rate of 64.2% and mean return of 0.551, marginally below the configuration's random baseline of 64.5%. This result reflects the challenge of converting stable loss reduction into a consistently better policy under three simultaneous sources of distributional shift.

3.2.2. Dueling DQN Training & Convergence

Dueling DQN follows a broadly similar training trajectory to DQN, with persistent oscillation and no clear convergence signal across 10,000 episodes. However, its final greedy policy achieves a survival rate of 68.0% and mean return of 0.639 - 3.5 percentage points above the random baseline and the strongest result in Config B.

The TD loss (Figure 10) shows a sharper initial spike than DQN, followed by a plateau around 0.02 that remains stable for the remainder of training, indicating that the network converges in its loss but not in its policy - consistent with the oscillating survival rates observed throughout. Figure 11 shows the evolution of the value and advantage stream magnitudes across training. The value stream dominates throughout, stabilising at approximately 0.8 after an initial peak, while the advantage stream remains consistently small at roughly 0.15. This separation confirms that the decomposition is functioning as intended: the network has learned to attribute most of the Q-value to patient state severity rather than to the marginal benefit of individual actions. Clinically, this aligns with the expectation that in sepsis, many patient trajectories are driven primarily by the severity of the underlying condition rather than by which specific treatment combination the agent selects at each step.

While the decomposed architecture adds parameters that require more data to train effectively, the 10,000 episode budget proved sufficient for the value stream to develop stable estimates, which likely explains its advantage over DQN at this training scale.

3.2.3. Config B Comparative Analysis

Table 3 summarises the final performance of both algorithms against the random baseline.

Algorithm	Mean Return	Survival Rate (%)	Avg Episode Length
Random baseline	0.549	64.5	-
DQN	0.551	64.2	9.9
Dueling DQN	0.639	68.0	8.8

Table 3 - Config B performance metrics

Dueling DQN outperforms DQN by 3.8 percentage points in survival rate and achieves a meaningfully higher mean return (0.639 vs 0.551), establishing it as the stronger Config B agent. DQN falls marginally below the random baseline (64.2% vs 64.5%), suggesting that the standard Q-network architecture does not benefit from additional training beyond a point and may be overfitting to the noisy wrapper-corrupted observations. The shorter mean episode length of Dueling DQN (8.8 vs 9.9) indicates earlier episode termination on average — consistent with a more decisive policy that commits to treatment choices rather than exploring, which in this case translates to higher survival.

Dueling DQN's advantage is attributable to its decomposed architecture: the value stream $V(s)$ provides a stable estimate of patient prognosis that is updated by all actions simultaneously, making it more resilient to the observation corruption introduced by the wrappers. DQN's single Q-stream must learn the full state-action mapping from scratch, which is more sensitive to the distributional shift across episodes.

Neither agent achieves substantial improvement over random behaviour — Dueling DQN improves by 3.5 percentage points while DQN falls slightly below baseline — underscoring that Config B is considerably harder than Config A. The combination of continuous observations, sparse rewards, and three simultaneous distributional shifts demands considerably more than 10,000 training episodes to produce reliably converged policies.

3.3. Config A vs Config B

A direct numerical comparison between configurations is not methodologically appropriate: these present fundamentally different problems, differing in state representation, model availability, and environmental fidelity. The raw survival rates should not be interpreted as evidence that tabular methods are superior to deep RL in general.

What the comparison does reveal is the cumulative cost of increasing problem complexity. Table 4 places all results side by side.

Algorithm	Config	Survival Rate (%)	vs. Baseline
Policy Iteration	A	78.8	+8.5pp
Q-Learning		68.5	-1.8pp
DQN	B	64.2	-0.3pp
Dueling DQN		68.0	+3.5pp

Table 4 - Config A vs. Config B performance metrics

Moving from Config A to B introduces three compounding sources of difficulty. Primarily, the loss of the transition model removes the possibility of exact Bellman evaluation, forcing both agents to approximate Q^* from sampled experience. The shift from discrete to continuous observations means the agent never revisits the same state exactly, eliminating the simple lookup that made tabular Q-Learning tractable.

Finally, the three clinical wrappers inject systematic noise and irreducible failure modes that no algorithm can fully compensate for: AcuteEventEnv in particular introduces patient deterioration independent of treatment, placing a ceiling on achievable survival rate regardless of policy quality.

Notably, the best agent from each configuration achieves similar survival rates — Q-Learning at 68.5% and Dueling DQN at 68.0% — despite the substantially greater complexity of Config B. This near-equivalence should not be mistaken for equal capability: Q-Learning operates on a fully known, noise-free MDP with 716 discrete states, while Dueling DQN navigates a continuous 47-dimensional observation space under three simultaneous failure modes. That both reach similar performance under such different conditions speaks to the architectural advantages of the dueling decomposition rather than to any equivalence of the problems. The gap between the ceiling of each configuration (10.8 percentage points) represents the honest cost of deploying RL in a setting that more faithfully resembles clinical reality. Closing this gap would require longer training, wrapper-aware regularisation, or reward shaping to provide a denser learning signal.

3.4. Creative Extension — Clinical Interpretability

3.4.1. Motivation

A survival rate alone is insufficient to validate a clinical RL agent. Two agents achieving similar survival rates may do so through fundamentally different treatment strategies: one through clinically appropriate moderate intervention, another through systematic under/overtreatment. Either extreme violates the foundational clinical principle of *primum non nocere* and would represent a risk in real deployment regardless of its headline survival figure.

The 25 discrete actions in this environment allow us to delve into what treatment strategies each agent really learned, which it favours, which it avoids, and whether those preferences align with best patient outcomes.

3.4.2. Treatment Strategy Analysis

All analyses are conducted over 1,000 evaluation episodes using a fixed greedy policy. Action frequency is computed as the proportion of total steps in which each cell of the vasopressor-fluid grid was selected. Survival rate per action is computed as the proportion of episodes in which the patient survived, conditioned on that action having been taken at least once during the episode. Figure 13 reveals fundamentally different treatment strategies across the three agents.

The random baseline distributes actions uniformly across the grid (~4% per cell), providing a flat reference.

DQN develops a more concentrated but broadly distributed strategy, with the highest preferences for zero-fluid combinations: Vaso 0/Fluid 0 (7.5%), Vaso 2/Fluid 0 (7.7%), and Vaso 3/Fluid 1 (7.2%). These preferences suggest DQN has learned to favour vasopressor administration with minimal IV fluid — a strategy that reduces overhydration risk at the cost of potentially undertreating fluid deficit.

Dueling DQN's strategy is strikingly different and clinically notable: 44.00% of all treatment steps select Vaso 0/Fluid 0 — complete withholding of both interventions — with a secondary concentration at Vaso 1/Fluid 0 (9.72%). Together, these two cells account for over 53% of all actions. This extreme conservatism is consistent with the dominance of the value stream observed in Figure 11: if the agent has learned that patient state severity $V(s)$ accounts for ~80% of the Q-value — with treatment advantage $A(s,a)$ contributing only ~15% — it effectively behaves as though intervention rarely changes outcomes. Clinically, this could reflect the agent internalising the effect of `AcuteEventEnv`: some episodes end in death regardless of treatment, which may have suppressed the agent's confidence in active intervention across the board.

Figure 14 presents the outcome associated with each cell for DQN and Dueling DQN respectively. For DQN, moderate vasopressor levels (Vaso 2) produce the highest survival rates across all fluid levels (74.8–75.7%), consistent with clinical guidelines that recommend moderate vasopressor support for haemodynamic stabilisation without excessive vasoconstriction. The worst outcomes for DQN occur at Vaso 1/Fluid 0 (55.8%) and Vaso 0/Fluid 0 (61.9%), precisely the cells it selects most frequently. This misalignment between action preference and outcome suggests DQN has not fully converged to an outcome-optimal policy — it concentrates on low-intensity actions that its Q-values overestimate. For Dueling DQN, the highest survival is associated with Vaso 0/Fluid 4 (84.3%) and Vaso 3/Fluid 3 (79.4%), neither of which is meaningfully utilised (0.57% and 0.70% respectively).

The agent's dominant action, Vaso 0/Fluid 0, yields a survival rate of 70.6% — above the random baseline but well below the best available cells. This represents a clinically suboptimal strategy: the agent has learned a conservative default that performs adequately on average but systematically misses higher-survival treatment combinations.

3.4.3. Findings & Clinical Implications

Three findings emerge from this analysis. First, both agents largely avoid the most aggressive treatment combinations (Vaso 4/Fluid 4), contrasting with the random baseline which assigns equal probability to all intensities. This avoidance of extreme interventions is clinically appropriate and suggests both agents have absorbed the intensity penalty signal, even if only weakly. Second, neither agent fully aligns its action preferences with the highest-survival cells. DQN concentrates on low-fluid combinations where survival is suboptimal; Dueling DQN concentrates overwhelmingly on complete non-intervention despite better outcomes being available elsewhere.

This misalignment is a direct consequence of incomplete training - the Q-values have not converged sufficiently for the greedy policy to consistently select the empirically best actions. Third, and most importantly, this analysis demonstrates that survival rate alone is an insufficient evaluation metric for clinical RL agents. DQN (64.2% survival) and Dueling DQN (68.0% survival) differ not only in aggregate performance but in fundamentally different treatment philosophies - one distributing interventions moderately, the other withholding treatment in the majority of encounters. A deployment decision based purely on survival statistics would obscure this distinction entirely. Interpretability analysis of the kind presented here is therefore a necessary complement to performance metrics in any clinical RL evaluation, and a prerequisite for the kind of clinical trust that would be required before deploying such agents in real ICU settings.

4. Visualizations

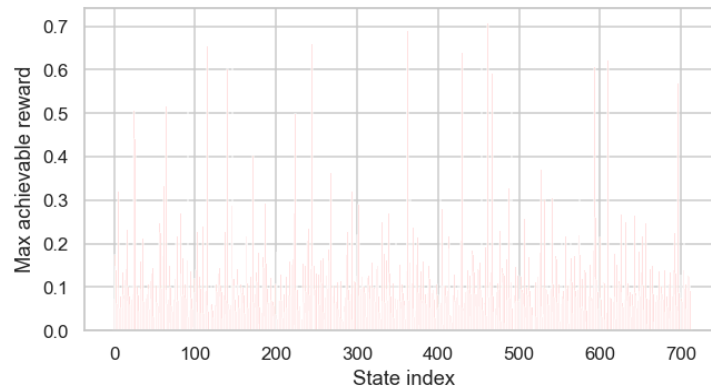


Figure 1 - Config A: Reward Sparsity: $\max_a R(s,a)$ across all states

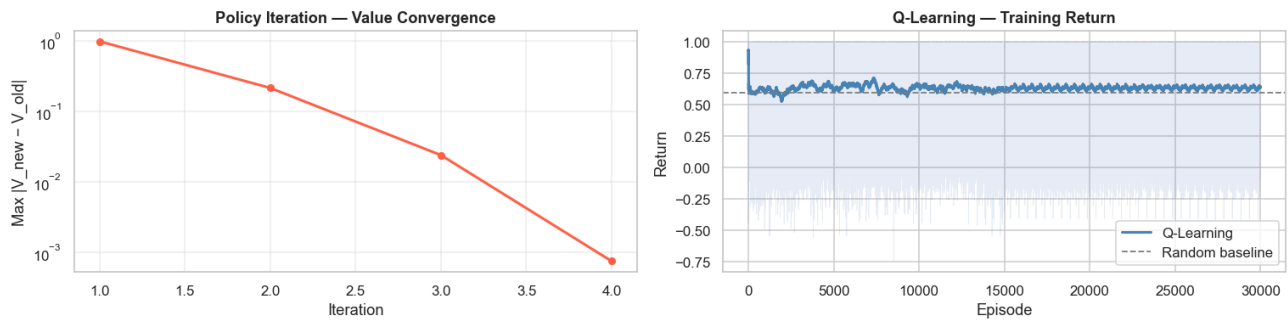


Figure 2 - Config A: Policy Iteration value convergence and Q-Learning training return.

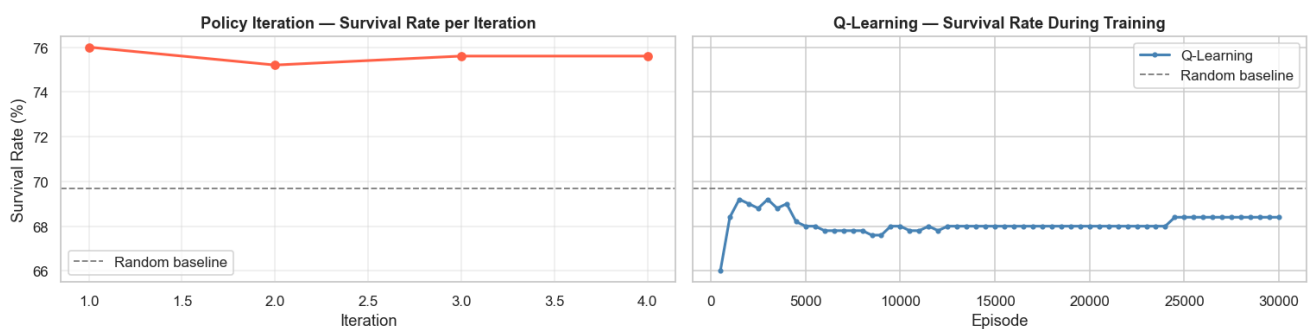


Figure 3 - Config A: Survival rate comparison across training - Policy Iteration per iteration and Q-Learning per episode (shared y-axis).

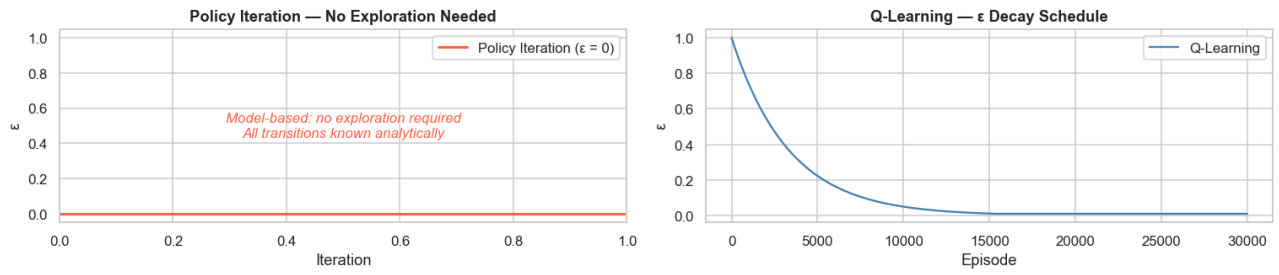


Figure 4 - Config A: Exploration vs Exploitation - Policy Iteration (left) and Q-Learning ϵ decay (right)

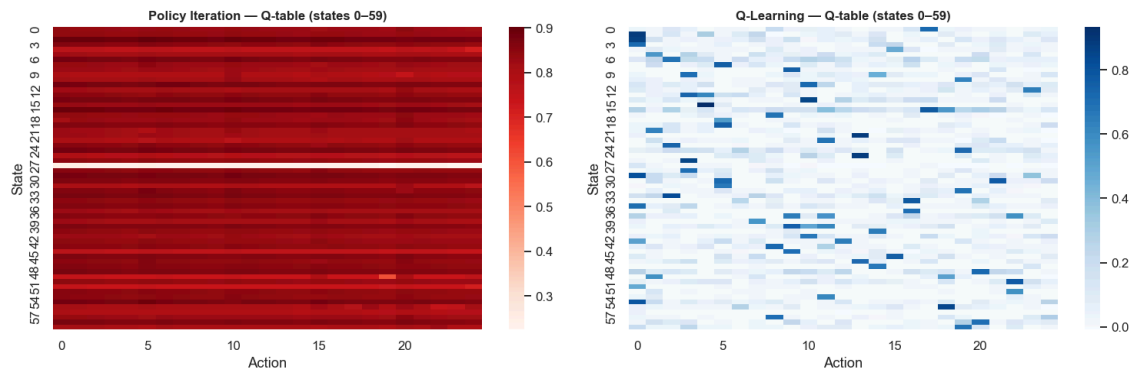


Figure 5 - Config A: Q-table Comparison - Policy Iteration (left) and Q-Learning (right), states 0–59

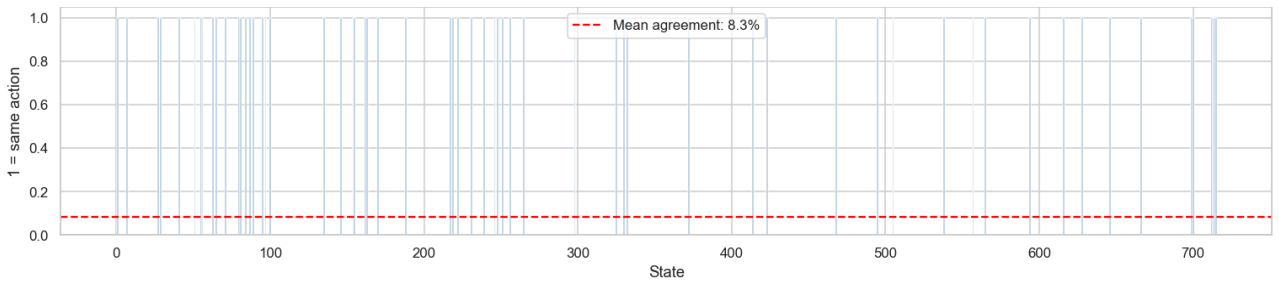


Figure 6 - Config A: Greedy policy agreement between Policy Iteration and Q-Learning across all 714 non-terminal states. Mean agreement: 8.3%.

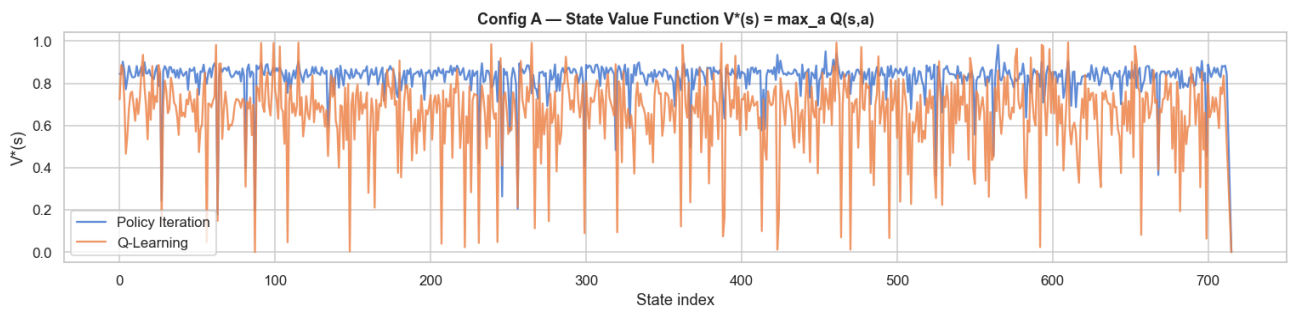


Figure 7 - Config A: State value function $V^*(s) = \max_a Q(s,a)$ for Policy Iteration and Q-Learning.

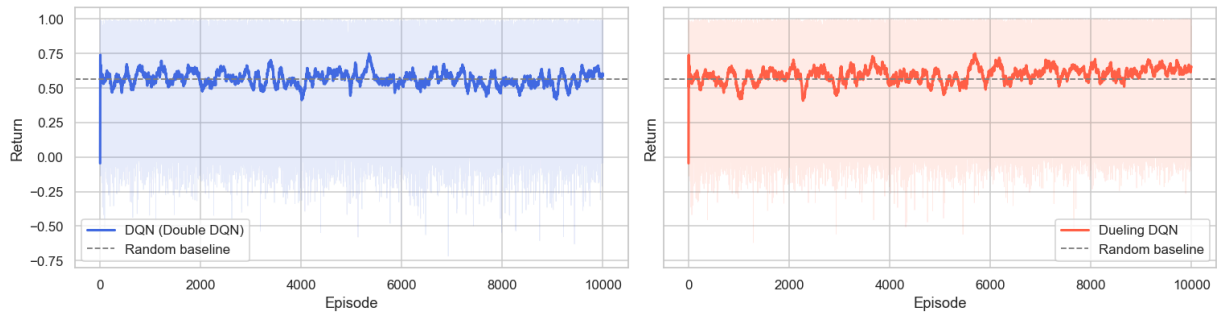


Figure 8 - Config B: DQN and Dueling DQN training return

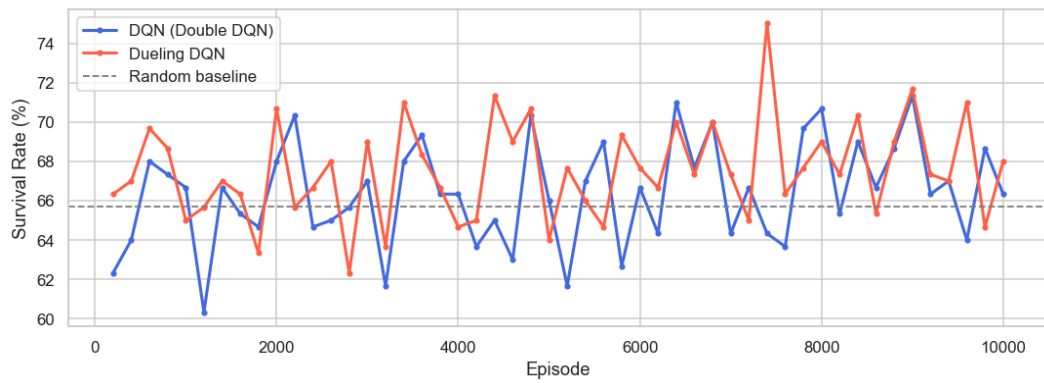


Figure 9 - Config B: Survival rate during training for DQN and Dueling DQN against the random baseline.

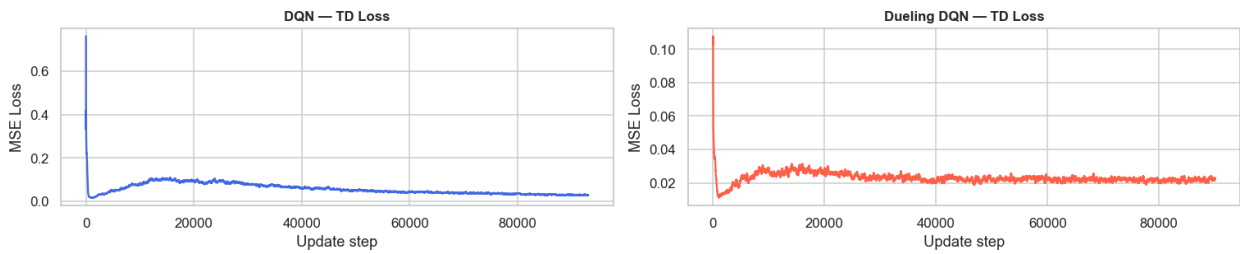


Figure 10 - Config B: TD loss curves for DQN and Dueling DQN over 10,000 update steps.

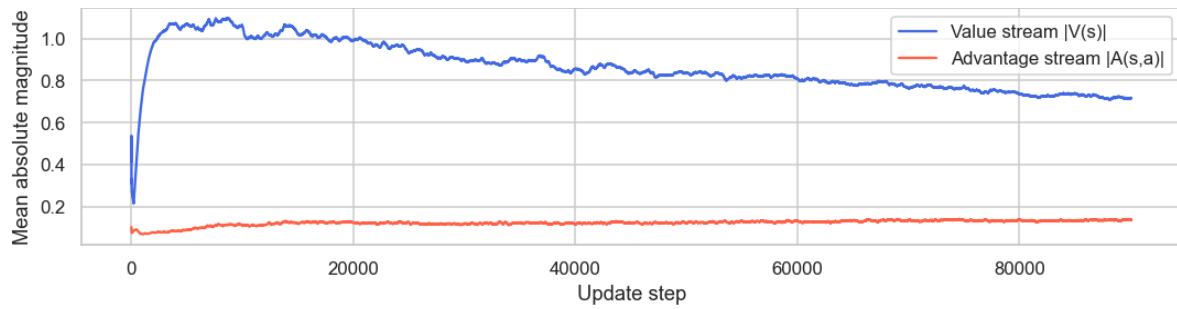


Figure 11 - Config B: Dueling DQN value and advantage streams magnitudes over training.

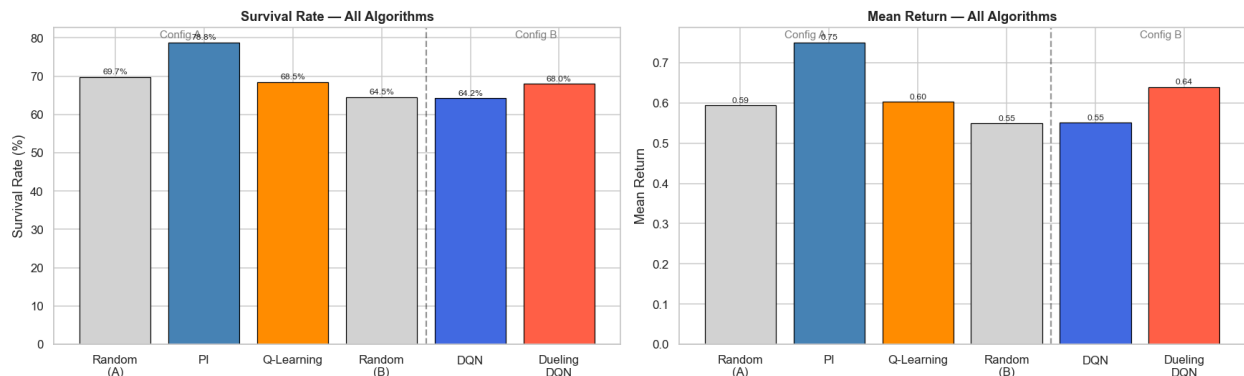


Figure 12 - Config A vs. Config B - Survival rate and mean return comparison

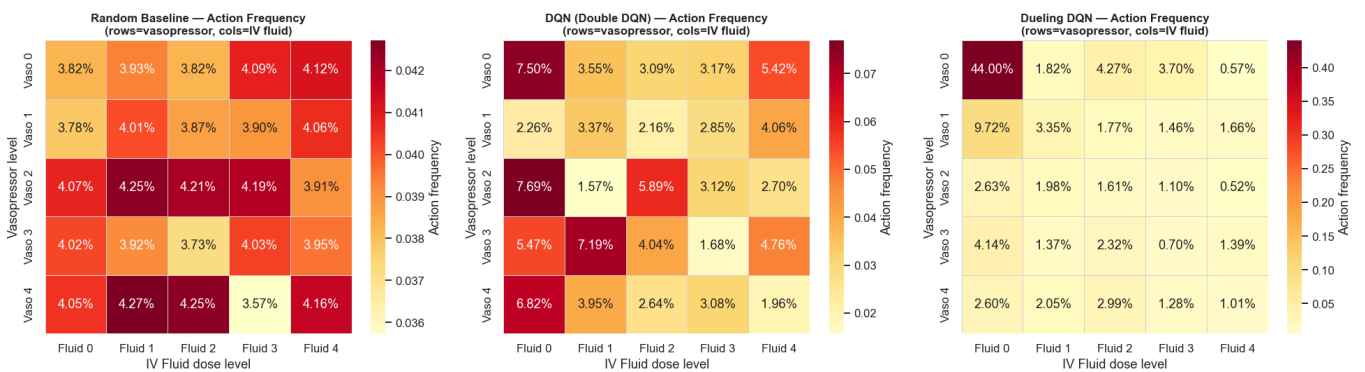


Figure 13 - Creative Extension: Action frequency across the vasopressor–fluid grid for Random Baseline, DQN, and Dueling DQN.

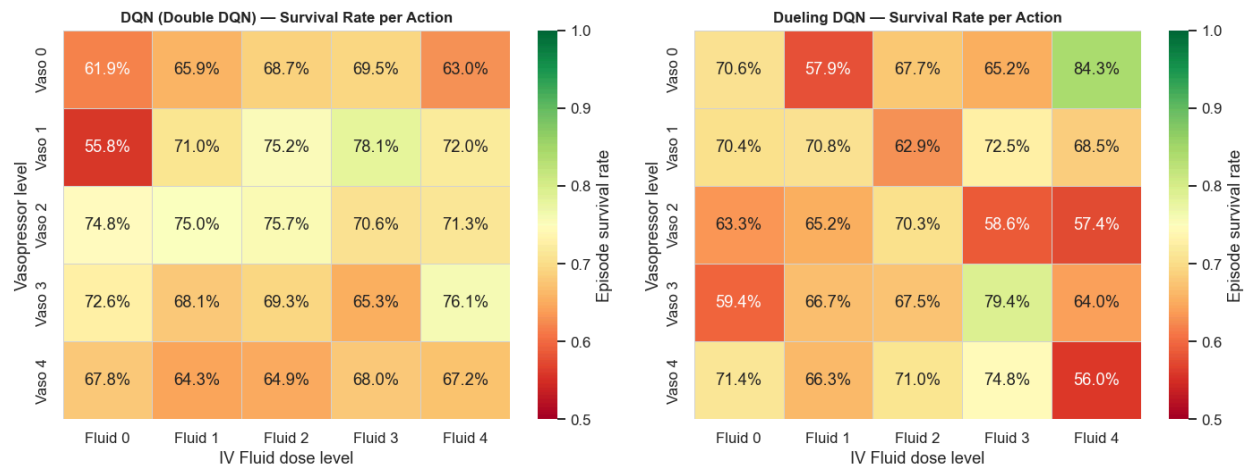


Figure 14 - Creative Extension: Episode survival rate per action cell for DQN and Dueling DQN.

5. Conclusion

5.1 Summary of Findings

This project evaluated four RL algorithms across two configurations of the ICU-Sepsis-v2 environment, with a creative extension examining the clinical interpretability of learned treatment strategies.

In Config A, Policy Iteration achieved 78.8% survival in 4 iterations by exploiting the known MDP model through exact Bellman evaluation. Q-Learning, operating without the model, required 30,000 episodes to converge and still reached only 68.5%, with just 8.3% policy agreement with the optimal PI policy. This gap quantifies the precise cost of model-free learning under sparse rewards: it is not a failure of the algorithm, but an unavoidable consequence of depending only on interactions to learn.

In Config B, the loss of the model, the shift to observations of continuous nature, and three simultaneous clinical wrappers substantially reduced achievable performance. Dueling DQN (68.0%) outperformed standard DQN (64.2%), with its decomposed architecture proving more resilient to observation corruption - the dominant value stream providing a stable prognosis estimate that DQN's single Q-stream could not replicate under noisy conditions. Neither agent achieved reliable convergence within 10,000 episodes.

The clinical interpretability analysis revealed that survival rate alone masks fundamentally different treatment behaviours. DQN distributes actions broadly with a preference for low-fluid combinations; Dueling DQN withholds all treatment in 44% of encounters. Both strategies diverge significantly from the highest-survival action cells, confirming that incomplete training produces policies that seem adequate on aggregate, but clinically suboptimal at the individual level.

5.2 Limitations

Several limitations constrain the conclusions that can be drawn. Both Config B agents were trained for only 10,000 episodes - insufficient for stable convergence under three simultaneous distributional shifts, as evidenced by persistent survival rate oscillation throughout training. The sparse, terminal reward signal slows credit assignment for all model-free methods, disproportionately affecting states visited infrequently.

Both deep RL agents share identical hyperparameters, which were not tuned for the Dueling architecture specifically. Finally, the evaluation metric (episode survival rate) does not capture treatment intensity or intermediate patient outcomes, limiting the clinical validity of the performance comparison.

5.3 Future Work

The most direct improvement would be to extend Config B training to significantly more episodes, which based on the training curves would likely allow both agents to move past the oscillation phase and converge more reliably. A simple reward shaping modification, adding a small positive signal when physiological indicators stabilize rather than waiting for the terminal outcome, would make the learning signal denser and easier to learn from.

Finally, running a basic hyperparameter search (varying learning rate, batch size separately for DQN and Dueling DQN) would allow a fairer architectural comparison than the shared configuration used here.